Q1.

At school Ellen heated some copper powder until it went black.

(a) Give the name of the black substance formed when copper reacts with oxygen.

.....

1 mark

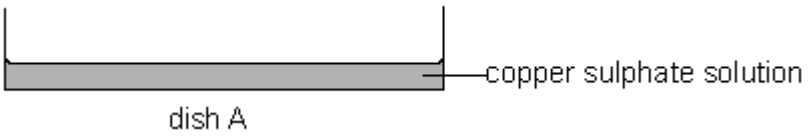
(b) Ellen added the black substance to some dilute sulphuric acid. The black substance reacted with sulphuric acid forming a blue solution of copper sulphate.

What type of substance is copper sulphate?
Tick the correct box.

an acid	☐	a compound	☐
an element	☐	a mixture	☐

1 mark

(c) (i) Ellen poured 20 cm^3 of the blue copper sulphate solution into a dish, A, as shown below.



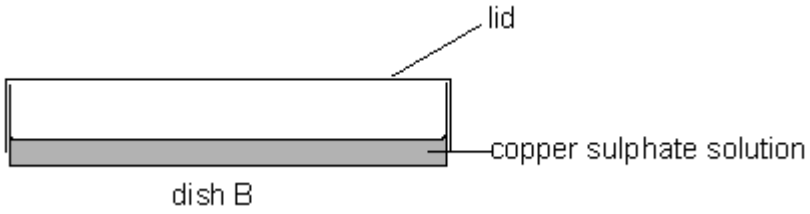
She left the dish in a room at 21°C for two days.
What **two** changes would Ellen observe in dish A two days later?

1. .....

2. .....

2 marks

(ii) Ellen poured 20 cm^3 of the same blue copper sulphate solution into another dish, B. She put a lid on dish B and left it in the room at 21°C for two days.



After two days the contents of dish B looked different from the contents of dish A. Give **one** difference Ellen would observe and explain how the lid caused this difference.

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.....

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2 marks
Maximum 6 marks

Q1.

(a) copper oxide

1 (L6)

(b) compound ✓

if more than one box is ticked, award no mark

1 (L5)

(c) (i) less solution **or** liquid **or** water

answers may be in either order

accept 'no liquid' ***or*** *'the liquid had gone'*
or *'liquid had evaporated'*

1 (L5)

crystals formed

accept 'solid copper sulphate formed'
or *'a blue solid appeared'*

1 (L5)

(ii) any **one** from

- there would be more of the solution left
  accept 'the solution would still be there'
- there would be fewer crystals
  accept 'no crystals'
  accept 'no change in dish B'

1 (L5)

any **one** from

- it prevented the evaporation
  accept 'it stopped the water leaving the dish'
- less water would have evaporated **or** gone

1 (L5)

[6]